BLACK PEN Answer all questions first. **GREEN PEN** Then tick correct answers; rewrite mistakes.

Flowcharts, Pseudocode and Programming Constructs

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: Answer all four before looking at the model answers.

1. Define abstraction.

2. Define decomposition.

3. Identify IPO for an image-file size calculation.

4. Explain why a journey-time algorithm should reject speed = 0.

Today's objectives

Recognise sequence, selection and iteration. Identify flowchart shapes and trace branches.

Key vocabulary - Read, cover, recall

sequence
Executing instructions in a stated order.

selection
Choosing a branch based on a condition.

iteration
Repeating instructions.

decision diamond
Flowchart shape containing a condition with branches.

Write one definition in your own words:

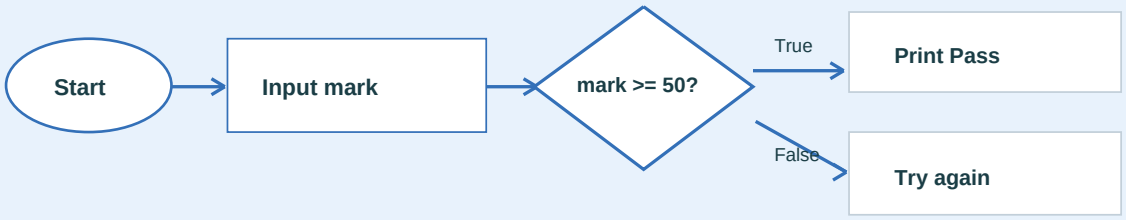
Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New learning - 2.1.2 / 2.2.1

Sequence	Instructions run in order; arrows on a flowchart show control flow.
Selection	A decision tests a condition and follows one of the branches, often using IF/ELSE.
Iteration	Instructions repeat; a loop can be count-controlled or condition-controlled.
Flowchart symbols	Use terminators (start/end), parallelograms (input/output), rectangles (processes), diamonds (decisions) and arrows. Pseudocode expresses the logic as text.



Sequence follows arrows; the diamond chooses a branch.

I Do - Worked example

Design a pass/fail algorithm: mark 52 passes when mark >= 50.

1. Start by reading a mark.
2. Use a diamond decision: is mark >= 50?
3. True branch prints Pass; false branch prints Try again.
4. Both branches lead to End. Pseudocode uses IF, ELSE and ENDIF.
5. For mark 52, the true branch prints Pass.

We Do - Guided practice

Design a flowchart to check whether an age is at least 18.

Use the model. Write your answer in BLACK PEN:

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Think and write

Code First - Predict > Write > Test > Improve

Read the code before you write or test anything.

```
fileGB = 12
freeGB = 10
if fileGB <= freeGB then
  print("Fits")
else
  print("Too large")
endif
```

Predict which branch runs when checking storage capacity.

1. Predict the output.

2. Change fileGB to 8.

3. Identify the decision condition and name one other factor in choosing storage.

You Do - Independent exam questions

Q1. Name two flowchart symbols and their functions. [2 marks]

Q2. Trace: x = 3; x = x + 4; print(x). Give the output and describe sequence. [3 marks]

Q3. Explain the different purposes of selection and iteration in a password-length checker which asks again if too short. [4 marks]

Exit Ticket - True or False?

Answer each statement before turning to the answers.

1. An IF statement automatically repeats until the condition changes. ☐ T ☐ F

2. A decision is usually shown with a diamond. ☐ T ☐ F

3. Iteration means repeating a set of instructions. ☐ T ☐ F

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Review & improve

GREEN PEN: Tick what is correct, then use the model to fix mistakes.
Do not just copy answers; explain what you changed.

Do Now - Answers

Q1 Keep important information; ignore irrelevant details. ☐

Q2 Split a task into smaller parts. ☐

Q3 Inputs width, height, depth; process multiply; output bits. ☐

Q4 Division by zero is undefined; validate positive speed before calculating. ☐

Code First - Model

Initially Too large is printed because 12 > 10. With fileGB = 8 the program prints Fits. A flowchart diamond would contain fileGB <= freeGB. Suitable storage also depends on speed, durability, portability, reliability and cost.

We Do - Model answer

Input age; decision diamond age >= 18; adult/under 18 outputs; end.

Exit Ticket - Check

1. FALSE - False: revise the new learning.
2. TRUE - True: the diamond contains a condition and branches.
3. TRUE - True: a loop repeats instructions.

You Do - Self-mark your exam answers

Q1 Diamond for a decision [1]; parallelogram for input/output [1] (other correct pairs accepted). ☐

Q2 Output 7 [1]; first assignment stores 3 [1]; next adds 4 before print [1]. ☐

Q3 Selection tests length [1] and branches by condition [1]; iteration repeats the input prompt [1] until an acceptable value is entered [1]. ☐

Try the question again

1. Copy the FULL question you got wrong (or found hardest):

2. In GREEN PEN, write a NEW complete answer. Show working: